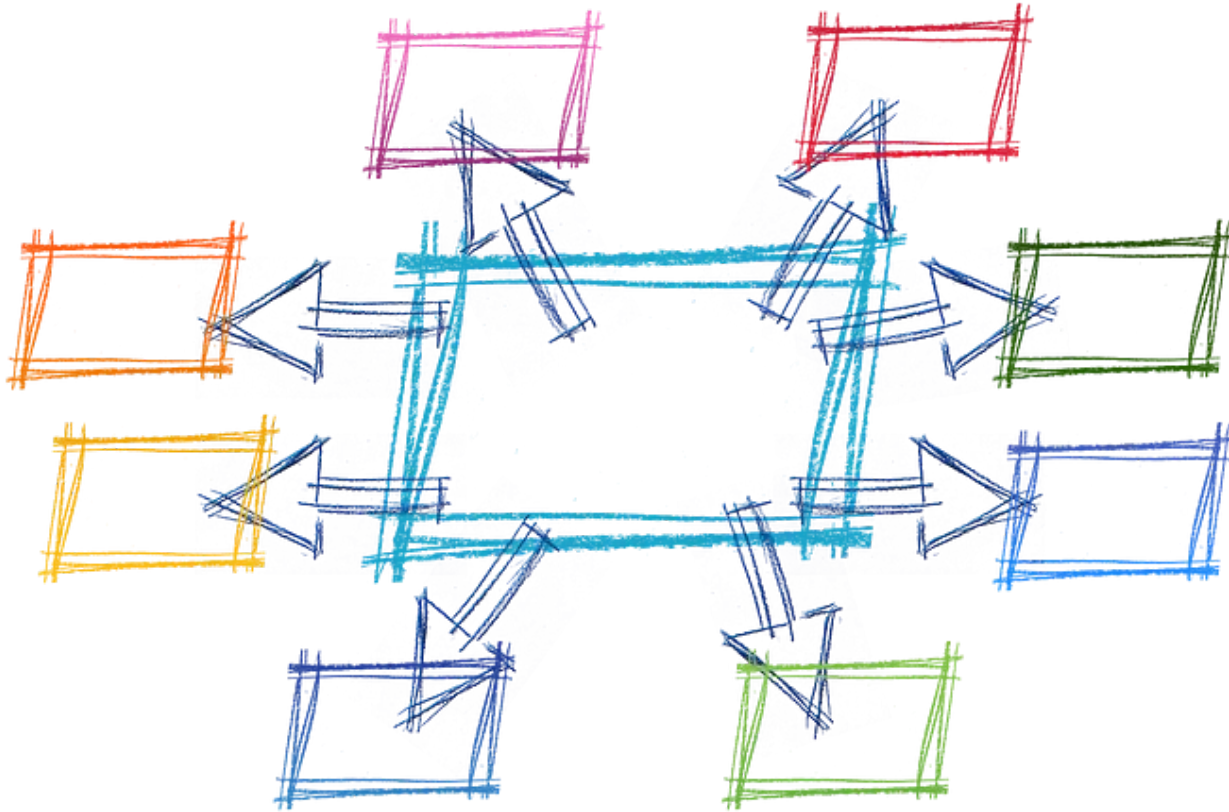




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Data Scientists, The 5 Graph Algorithms that you should know

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We as data scientists have gotten quite comfortable with Pandas or SQL or any other relational database.

We are used to seeing our users in rows with their attributes as columns. But does the real world really behave like that?

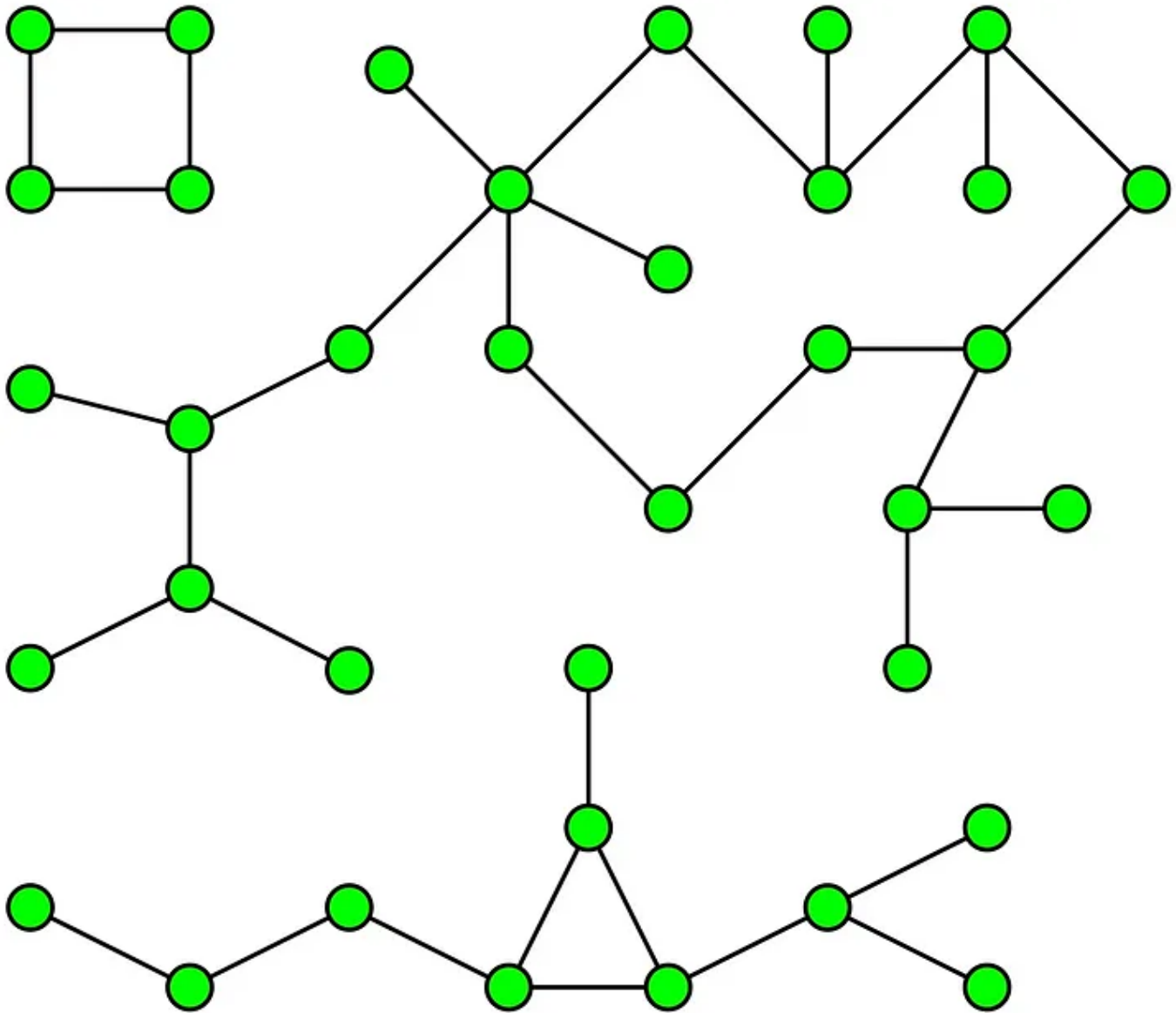
In a connected world, users cannot be considered as independent entities. They have got certain relationships between each other and we would sometimes like to include such relationships while building our machine learning models.

Now while in a relational database, we cannot use such relations between different rows(users), in a graph database it is fairly trivial to do that.

In this post, I am going to be talking about some of the most important graph algorithms you should know and how to implement them using Python.

Also, here is a [Graph Analytics for Big Data](#) course on Coursera by [UCSanDiego](#) which I highly recommend to learn the basics of graph theory.

1. Connected Components



A graph with 3 connected components

We all know how clustering works?

You can think of Connected Components in very layman's terms as a sort of a hard clustering algorithm which finds clusters/islands in related/connected data.

As a concrete example: Say you have data about roads joining any two cities in the world. And you need to find out all the continents in the world and which city they contain.

How will you achieve that? Come on give some thought.

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Written by Rahul Agarwal

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